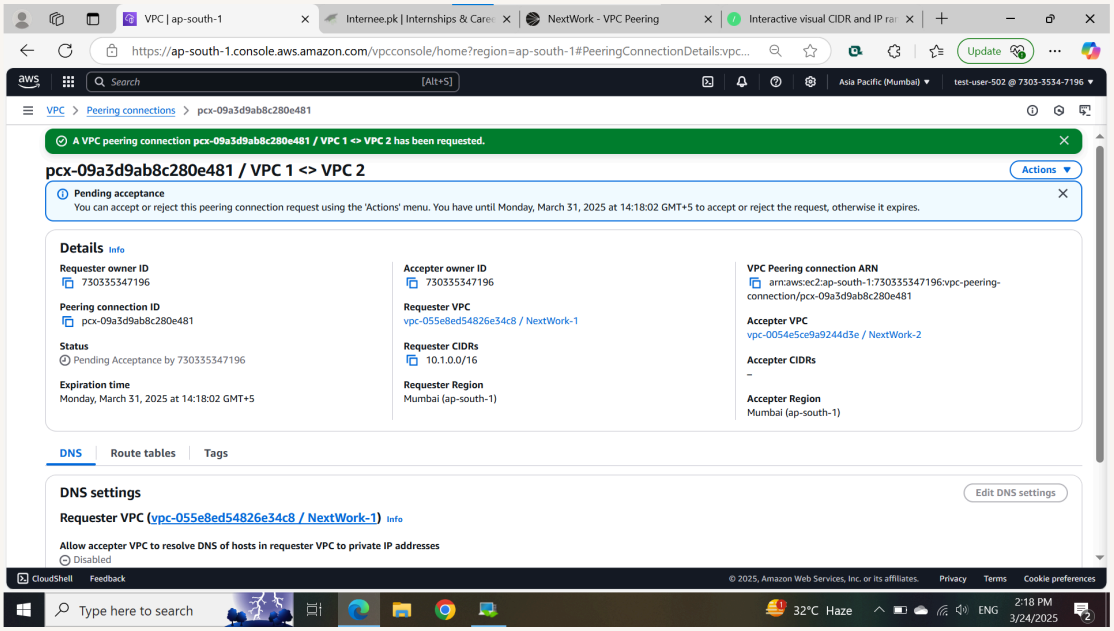
VPC Peering

zubair Hashmat



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Introducing Today's Project!

What is Amazon VPC?

AWS's foundational networking service that lets me create my own networks,control traffic flow and security and organize my resources into public/private subnets.

How I used Amazon VPC in this project

I used Amazon VPC to set up a multi-VPC architecture (I set up two VPCs), create peering connection between them and update security group rules to run a successful connectivity test to validate my VPC peering set up.

One thing I didn't expect in this project was...

I didn't expect to need a public IPv4 address for EC2 Instance Connect to work. Also I didn't expect that Elastic IPs can assign static public IPv4 addresses to resources.

This project took me...

The project took me 2 hours to complete.

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In the first part of my project...

Step 1 - Set up my VPC

In this step, we are using the VPC resource map/launch wizard to create two VPCs and their components in just minutes.

Step 2 - Create a Peering Connection

I am setting up a VPC Peering Connection, which is a VPC component designed to directly connect two VPCs together.

Step 3 - Update Route Tables

I am editing route tables for VPC1 and VPC2, so that traffic can be directed to the peering connection set up.

Step 4 - Launch EC2 Instances

I am launching an EC2 instance in each of the VPCs(VPC1 and VPC2), so that I can directly connect with my instances later and test my VPC peering connection.

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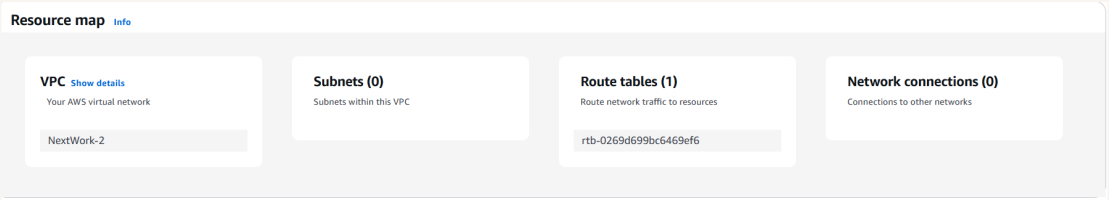
Multi-VPC Architecture

I started my project by launching two VPCs - They have unique CIDR blocks and they each have one public subnet.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16 respectively.They have to be unique because once you set up a VPC peering connection, route tables need unique addresses for correct routing across VPCs.

I also launched 2 EC2 instances

I did not set up key pairs for these EC2 instances as I am using EC2 instance Connect to directly connect with my EC2 instance later in this project, which handles key pair creation and management for us.



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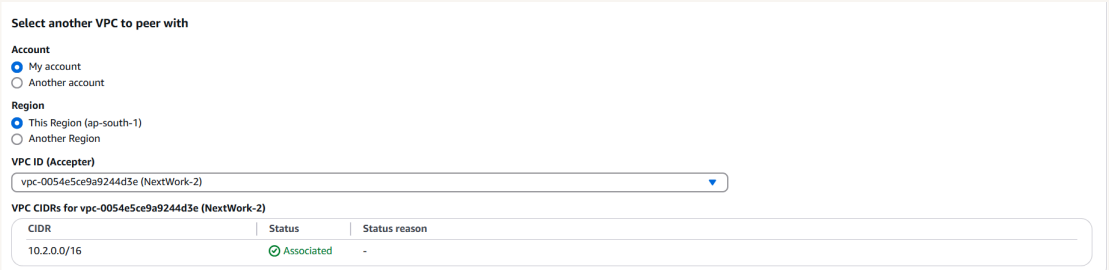
VPC Peering

A VPC peering connection is a direct connection between two VPCs.A peering connection lets VPCs and their resources route traffic between them using their private IP addresses. This means data can now be transferred between VPCs.

VPCs would use peering connections to establish a secure and private network link between two VPCs, enabling seamless communication between resources in both.

In VPC peering, the Requester is the VPC that initiates a peering connection. The Accepter is the VPC that receives a peering connection request! The Accepter can either accept or decline the invitation.



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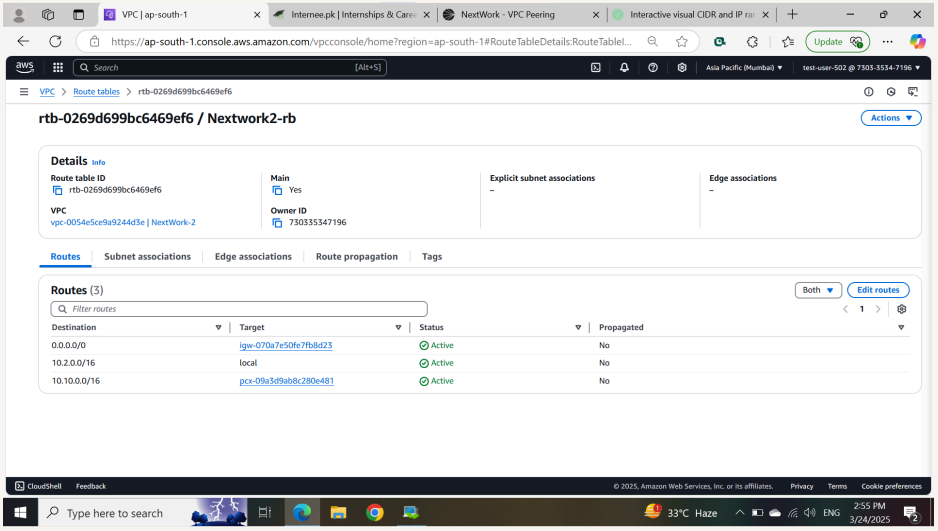
Updating route tables

My VPCs' route tables need to be updated because the default route table does not have a route using the peering connection yet - this needs to be set up so that resources can be directed to peering connection when trying to reach other VPC.

My VPCs' new routes have a destination of the other VPC's CIDR block.The routes' target was the peering connection I set up.



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In the second part of my project...

Step 5 - Use EC2 Instance Connect

I am using EC2 instance Connect to connect directly with my first EC2 instance.I need to do this because I need to use my EC2 instance for connectivity tests later in this project.

Step 6 - Connect to EC2 Instance 1

In this step, I am going to use EC2 Instance Connect to connect to Instance 1 (one more time)! Fix (another) error if occur.

Step 7 - Test VPC Peering

I am going to use Instance - Nextwork VPC 1 to attempt a direct connection with Instance - Nextwork VPC 2 so that I can validate that my peering connection is set up properly.

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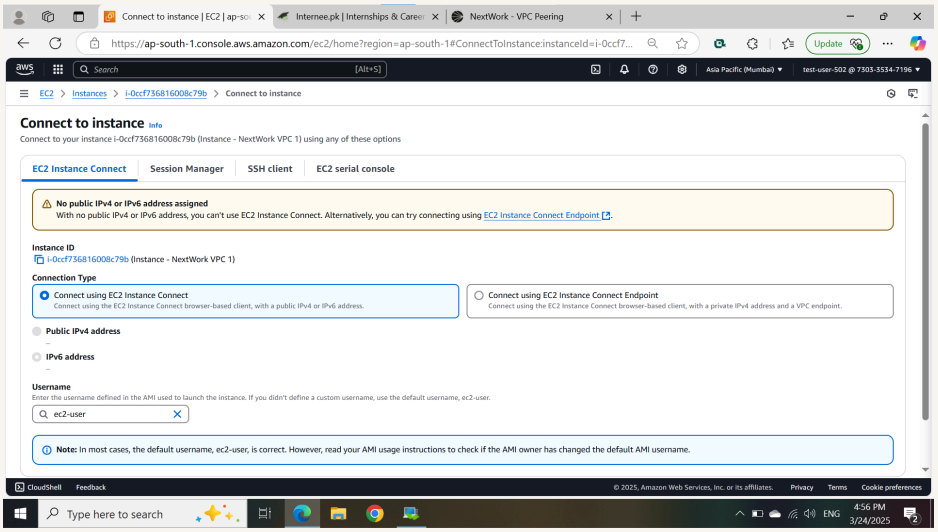
Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to directly connect with Instance - NextWork VPC 1 just by using the AWS Management Console.

I was stopped from using EC2 Instance Connect as my instance did not have public IPv4 address.In order for EC2 Instance Connect to work, the EC2 instance must have a public IPv4 address and be in a public subnet.



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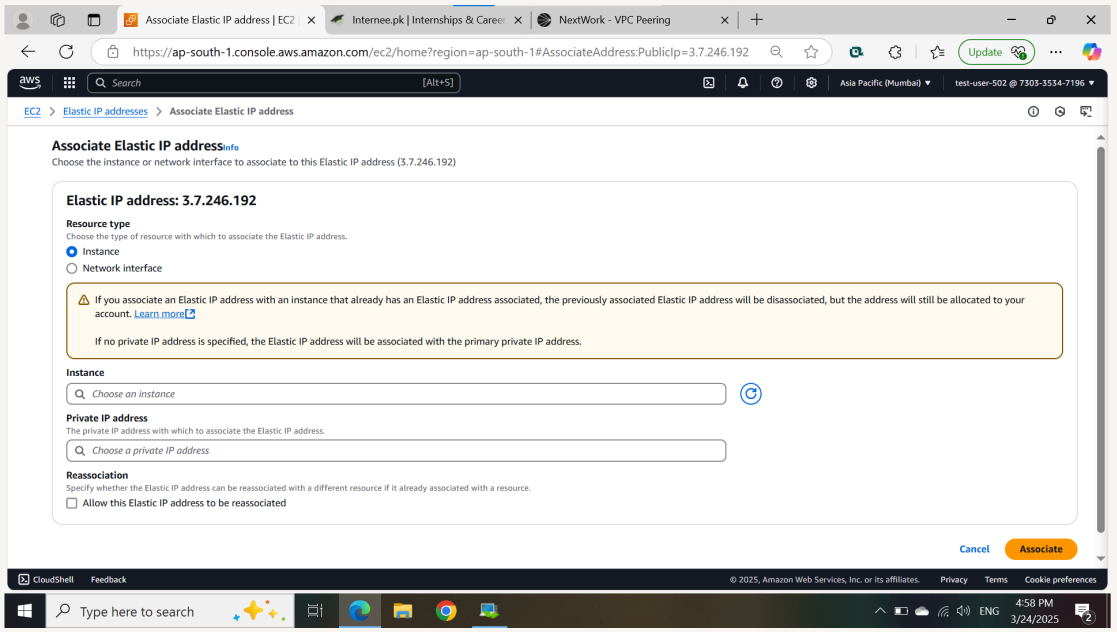
Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are static public IPv4 addresses that I can request for my AWS Account, and then delegate to specific resources.

Associating an Elastic IP address resolved the error because it gives my EC2 instance a public IP address, fulfilling the requirements for Instance Connect to work.



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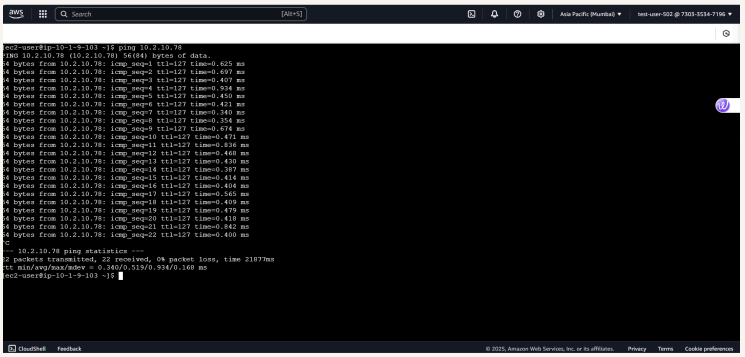
Troubleshooting ping issues

To test VPC peering,I ran the command ping 10.2.xxx.xxx i.e. the private IPV4 address of the other EC2 instance in VPC 2.

A successful ping test would validate my VPC peering connection because this ping test would not get any replies from the other EC2 instance if the peering connection did not successfully connect my VPCs.

I had to update my second EC2 instance's security group because it was not letting in ICMP traffic - which is the traffic type of a ping message.I added a new rule that allows ICMP traffic coming in from any resource.